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Comparative Evaluation of Rule-Based, PID, Model-Predictive, Reinforcement-Learning, and Hybrid Control Strategies for Residential HVAC Systems in a Composite Climate

A Year-Long Simulink-Based Simulation Study of a 3BHK Residence in Pune, India

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Abstract Residential air conditioning is the fastest-growing driver of electricity demand in Indian cities, and most homes still run on simple thermostats or hand-tuned PID logic that trades comfort for efficiency or the reverse. This paper presents a full-year (8760-hour), physics-based Simulink simulation of a representative three-bedroom, hall, kitchen (3BHK) residence in Pune, Maharashtra. Five controllers govern the same variable refrigerant flow (VRF) air conditioner, indirect evaporative cooler (IEC), and dedicated outdoor air system (DOAS): a reactive rule-based baseline, a three-zone anti-windup PID controller, a receding-horizon model-predictive controller (MPC), a deep Q-network (RL-DQN) reinforcement-learning controller, and a hybrid controller that fuses MPC and PID output through adaptive, confidence-weighted blending. The building is modeled as a three-zone lumped-capacitance thermal network coupled to a Pune typical-meteorological-year climate profile, a variable-COP VRF model, and a two-pollutant (CO_2, $PM_{2.5}$) indoor-air-quality model. Against the baseline (RMSE 6.84 °C, 11.3% comfortable hours, 7110.8 kWh/yr), all four advanced controllers cut tracking error by 68–75% and more than sextupled the comfortable-hour fraction. MPC and Hybrid tied for the tightest tracking at 1.73 °C RMSE, PID delivered the largest energy saving at 23.2% (5460.9 kWh), and Hybrid produced the best combined outcome (21.8% less energy, 68.1% comfortable hours). RL-DQN, run greedily from pretrained weights, was the least accurate of the four (2.17 °C RMSE, 57.8% comfortable hours). Indoor CO_2 stayed well controlled by the humidity-triggered DOAS. Indoor $PM_{2.5}$ levels remained high throughout the year, ranging from 160 to 195 $\mu g/m^3$. These concentrations were well above the WHO and Indian NAAQS limits, even though a HEPA H13 filter with a reported single-pass efficiency of 99.95% operated continuously. The main limitation was the filtration arrangement: the filter treated recirculated air, while outdoor $PM_{2.5}$ entering through infiltration remained unfiltered. This

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result shows that changing the HVAC control strategy alone cannot resolve poor indoor air quality when the filtration system is not designed to handle the pollution entering the building.

Keywords HVAC control, VRF systems, model-predictive control, reinforcement learning, PID control, hybrid control, thermal comfort, indoor air quality, $PM_{2.5}$, residential energy simulation, Simulink, Pune.

1 Introduction

Space-cooling electricity consumption in India increased by 21% between 2019 and 2022, accounting for nearly 10% of the country's total electricity demand [1]. This growth is linked to the rising adoption of room and split air conditioners, along with increasing exposure to urban heat. Pune, located in Maharashtra's composite climate zone, experiences distinct seasonal conditions. During the pre-monsoon period, outdoor temperatures can exceed 40 °C. The monsoon brings high humidity, often above 80%, while winters remain relatively mild and dry. A residential HVAC system must therefore handle several demands throughout the year, including sensible cooling during summer, latent-load control during the monsoon, and maintaining indoor air quality through adequate ventilation.

Most homes here still run on on/off thermostats or single-loop PID logic tuned for one nominal set point, with no regard for zone-to-zone thermal coupling, time-of-day (ToD) tariffs, or how the sensible cooling equipment (VRF), latent/evaporative assist (IEC), and fresh-air ventilation (DOAS) interact. Model-predictive control (MPC) and reinforcement learning (RL) have shown promise in commercial and simulated residential buildings. MPC exploits a forecast of thermal disturbances to anticipate cooling demand. RL learns a control policy directly from interaction with a building model. But full-year, physically grounded comparisons against simple baselines are still scarce in the Indian residential context, and almost nobody evaluates control strategy and indoor air quality together.

This paper closes part of that gap with an integrated, physics-based simulation of a real-world-representative 3BHK residence in Pune, built entirely in MATLAB/Simulink and run over a full 8760-hour year. Five controllers act on an identical building, equipment, and weather model: a reactive rule-based baseline, a three-zone anti-windup PID controller, a receding-horizon MPC controller, a deep Q-network (RL-DQN) controller in greedy inference mode, and a hybrid controller that fuses MPC and PID outputs through an adaptively weighted confidence mechanism with a safety layer. The contributions are:

- A coupled three-zone RC thermal network, variable-COP VRF model, indirect evaporative cooler, DOAS ventilation model, and a two-pollutant (CO_2, $PM_{2.5}$) IAQ model, all built as a single Simulink model.

- A full-year (8760 h), like-for-like comparison of five controllers spanning the range from static rules to learned policies, using an identical building, climate, and equipment envelope.
- A quantified comfort–energy–tracking trade-off, reported as RMSE, comfortable-hour fraction, annual energy, electricity cost under a ToD tariff, and embodied carbon.
- A joint evaluation of thermal control and indoor air quality that surfaces a filtration-architecture limitation invisible to thermal metrics alone.

Section 2 reviews the control paradigms compared here. Section 3 describes the building, equipment, air-quality, and control models, along with the simulation environment. Section 4 presents the results, Section 5 discusses their implications, Section 6 outlies the study's limitations, and Section 7 concludes the paper.

2 Background on HVAC Control Paradigms

PID control remains widely used in residential and commercial HVAC systems because it is simple, easy to understand, and computationally inexpensive. However, it responds to the current control error rather than anticipating future conditions. Its performance also depends on appropriate gain tuning, which becomes more challenging when a single control loop serves several thermally connected zones with different heating and cooling demands.

Model-predictive control (MPC) addresses this limitation by optimizing a sequence of control actions over a moving time horizon using a model of the building's thermal behaviour [2]. This approach is particularly useful for buildings with high thermal mass and in situations where time-of-day (ToD) electricity tariffs create opportunities to shift energy use [3]. However, MPC depends on the accuracy of its predictive model and requires more computation than conventional PID control. In this study, computational demands are reduced by using a simplified, low-order predictive model within the controller instead of the full physical building model.

Reinforcement learning, particularly value-based methods such as deep Q-networks (DQN) [6], offers another approach to HVAC control. These methods can learn control policies from data or simulation without requiring an explicit model of the system dynamics. Their application to building energy management and HVAC control has received considerable attention in recent years [20, 21].

In practice, RL policies are sensitive to state representation, reward shaping, and the range of conditions seen during training [20, 21]. A pretrained policy deployed greedily, with no further online adaptation, has been reported to underperform well-tuned classical controllers when the training regime does not fully capture the deployment environment [22, 23].

Hybrid architectures try to combine MPC's foresight with PID's robustness and simplicity, using fixed or adaptively learned blending weights plus a safety layer to guard against degenerate behavior from either sub-controller [7, 8].

The Hybrid controller in this paper follows that pattern: a rolling-window, inverse-error confidence weighting between an MPC-like predictive layer and a three-zone PID layer, plus a bounded proportional safety term.

3 Methodology

3.1 Case-Study Building and Climate

The case-study dwelling is a 95 m^2 3BHK residence in Pune, in composite climate zone ASHRAE 0B (warm-humid transitional). It is divided into three thermal zones of unequal size and orientation, each modeled as a lumped thermal capacitance coupled to its neighbors and to the outdoor environment.

- Zone 1, Master Bedroom: 30 m^2 (31.6% of conditioned floor area).
- Zone 2, Living + Dining: 45 m^2 (47.4%), the primary zone, used as reference for MPC, RL, and PID night-setback logic.
- Zone 3, Second Bedroom: 20 m^2 (21.1%).

All three zones have a ceiling height of 2.85 m, giving a total conditioned volume of 270.75 m^3. Heat transfer between adjacent zones is represented by inter-zone thermal resistances of $R_{12} = 0.0514$ K/W and $R_{23} = 0.0541$ K/W. Each zone's envelope conductance includes contributions from the walls, roof, glazing, and infiltration. Table 1 summarizes the lumped envelope conductances (UA) and thermal capacitances (C), which were derived from a 150 mm reinforced cement concrete (RCC) slab, masonry walls, and furnishings. The resulting zone thermal time constants, calculated as $\tau = C/UA$, range from 56 to 66 hours. These values represent the considerable thermal inertia associated with masonry residential buildings.

Outdoor conditions are taken from a typical meteorological year (TMY) profile representative of Pune. The profile includes dry-bulb temperature, relative humidity, global horizontal irradiance, wet-bulb temperature, and a seasonal index. A sol-air temperature correction is applied to account for roof heat gains, using an absorptance of 0.70, an emissivity of 0.90, and a longwave correction of $\Delta R = 63$ W/m^2). Solar and internal heat gains are also calculated separately for each zone.

3.2 HVAC Equipment Models

3.2.1 Variable Refrigerant Flow (VRF) Air Conditioner

The VRF unit has a rated capacity of 11.0 kW, with a minimum capacity of 2.2 kW and a maximum of 11.0 kW. Its reference coefficient of performance (COP) is 4.2 at an outdoor dry-bulb temperature of 35 °C , following the Indian Seasonal Energy Efficiency Ratio (ISEER) methodology [9]. The COP decreases approximately quadratically as outdoor temperature rises, using coefficients of 0.035 and 0.0003. This representation captures the decline in VRF

Table 1 Three-Zone Building Envelope and Thermal-Mass Parameters

Parameter	Zone 1	Zone 2	Zone 3
Floor area (m^2)	30.0	45.0	20.0
Area fraction	0.316	0.474	0.211
UA wall (W/K)	4.09	5.57	3.42
UA roof (W/K)	8.04	12.06	5.36
UA glazing (W/K)	17.56	23.94	14.68
UA infiltration (W/K)	14.10	21.14	9.40
UA total (W/K)	43.78	62.72	32.86
Capacitance C (MJ/K)	9.94	14.90	6.62
Time const. τ (h)	63.0	66.0	56.0
VRF capacity share	31.6%	47.4%	21.1%

efficiency at higher ambient temperatures reported in field studies. The unit operates over an outdoor temperature range of 15-48 °C, covering Pune's design conditions, including its documented summer peak of 41 °C.

3.2.2 Indirect Evaporative Cooler (IEC)

A supplementary indirect evaporative cooling (IEC) system is included, with 80% wet-bulb effectiveness, a 150 W auxiliary fan, and an airflow rate of 0.35 m^3/s. It operates when outdoor relative humidity falls below 50% and the dry-bulb temperature exceeds 28 °C. These conditions typically occur during Pune's dry summer and pre-monsoon periods, when evaporative cooling is most effective. The IEC system also requires the active controller's cooling command to exceed 200 W before it switches on. Its operation therefore depends on both outdoor conditions and cooling demand, meaning that annual runtime can differ between controllers. The supply-air temperature is estimated as the outdoor wet-bulb temperature plus 2.0 °C.

3.2.3 Dedicated Outdoor Air System (DOAS)

A dedicated outdoor air system (DOAS) supplies conditioned fresh air at 0.06 m^3/s when outdoor relative humidity exceeds 65% and the dry-bulb temperature exceeds 22 °C. These conditions occur mainly during the monsoon, when the system helps manage latent loads and dilute indoor CO2 in line with ASHRAE 62.1/62.2 ventilation guidance[10]. Unlike the IEC system, DOAS activation depends only on outdoor weather conditions, not on the cooling command from the controller. As a result, its runtime and annual energy consumption remain the same across all five control strategies. Baseline infiltration is modelled separately at 0.0376 m^3/s, equivalent to 0.5 air changes per hour (ACH). The IAQ subsystem uses a separately rounded value of 0.0375 m^3/s for the same quantity. Although these values differ by less than 0.1% and have no meaningful numerical effect, the discrepancy is documented here to support reproducibility.

3.3 Indoor Air Quality Model

The indoor air quality model tracks two pollutants, CO_2 and $PM_{2.5}$, using mass-balance equations that account for outdoor concentrations, indoor generation, ventilation, and infiltration. The $PM_{2.5}$ model also includes particle deposition and filtration. Both models assume a household of four occupants.

- CO_2: The outdoor background concentration is set to 420 ppm, with indoor generation scaled to four occupants. Infiltration and fresh air supplied by the DOAS dilute indoor CO_2. No mechanical CO_2 removal system is modelled, and the DOAS airflow is not included in the $PM_{2.5}$ balance described below.
- $PM_{2.5}$: The model uses a Pune-representative outdoor PM2.5 time series, with 30% of recirculated air passing through a HEPA H13 filter with a single-pass efficiency of $\eta = 99.95\%$. Particle removal through deposition is represented by a first-order loss rate of $\lambda = 1.39 \times 10^{-4}$ s^{1}, consistent with rates reported in residential $PM_{2.5}$ infiltration studies [11]. Indoor sources, including cooking and background activities, are also included. Outdoor $PM_{2.5}$ enters the model only through infiltration. The fresh-air supply from the DOAS is included in the CO_2 model but is excluded from the $PM_{2.5}$balance.

3.4 Control Strategies

All five controllers operate on the same three-zone building and the same VRF/IEC/DOAS equipment. The PID, MPC, and Hybrid controllers use a nominal setpoint of 24 °C with a 2 °C night setback between 23:00 and 06:00. The Baseline and RL-DQN controllers maintain a fixed 24 °C target throughout the day, without a night setback. This difference creates an asymmetry in the comparison and is discussed further in Section 6. Comfortable hours are defined using a Zone 2 temperature range of 22–26 °C. This range is close to the 23–26 °C air-conditioned comfort range recommended by India's National Building Code and the associated India Model for Adaptive Comfort[12,13].

3.4.1 Baseline (Reactive Rule-Based)

A single-zone-average threshold controller: it issues cooling only when the average zone temperature exceeds the set point, with duty fraction chosen by simple outdoor-condition rules. Like all five controllers here, it has no heating capability. Commanded cooling is clamped at zero, never negative, so that is not what sets the baseline apart. What is unique to the baseline is the complete absence of a deadband, anti-windup, or output rate limiter. A single hourly command can jump to near-maximum capacity whenever the instantaneous temperature error is moderate, and, as Section 4.1 shows, that produces a large single-step overshoot given the zone's finite thermal capacitance.

3.4.2 PID (Three-Zone, Anti-Windup)

Independent PID loops run for each zone ($K_p = [180, 300, 120]$ W/°C for Zones 1–3, $K_i = K_p/2400$, $K_d = K_p \times 120$, integrator limits $[3000, 5000, 2000]$, 0.4 °C deadband, conditional-integration anti-windup). Each zone's output capacity is capped at its floor-area fraction of the 11 kW VRF, and the three are summed into a single plant command.

3.4.3 MPC (Simplified Receding-Horizon)

A 12-step-ahead predictive controller uses a first-order discrete zone model (pole $a = 0.85$) driven by an outdoor-temperature disturbance estimate, an exponentially decaying (0.95^k) weighting of predicted overheating over the horizon, a hard 20–105% capacity band, an emergency override above set point +3.0 °C, and a ±30%/hour output rate limit. The implementation also computes a ToD-tariff weighting term, but that term never connects to the final demand calculation in the deployed code, so it has no effect on MPC's output. Unlike the Hybrid controller below, MPC's cooling command in this study is not actually tariff-aware, despite what the source code intends.

3.4.4 RL-DQN (Deep Q-Network, Greedy Inference)

A trained deep Q-network [6], following state-representation conventions used elsewhere in DRL-based whole-building HVAC control [14], maps an 8-feature normalized state (Zone-2 temperature error, outdoor temperature, relative humidity, sin/cos hour-of-day, season index, ToD tariff, and a secondary temperature-error term) through two 128-unit ReLU dense layers to Q-values over six discrete duty-fraction actions (0, 15, 30, 50, 75, 100% of 11 kW). The state features and no-action deadband reference a fixed 24 °C target only. Unlike PID, MPC, and Hybrid, RL-DQN's inputs do not encode the night-setback set point at all. The policy runs greedily ($\arg\max Q$, capped at 75%) from frozen, pretrained weights, with a ±1.0 °C no-action deadband and a ±35%/hour rate limit. No further learning happens during the evaluation year.

3.4.5 Hybrid (Adaptive MPC + PID + Safety Fusion)

A three-layer controller. Layer 1 is a 24-step MPC-like predictive term (pole $a = 0.67$, 0.93^k horizon weighting, ToD-tariff multiplier, staged overrides at 0.8/2.0/3.5/5.0 °C above set point, ±28%/hour rate limiting). Layer 2 is a three-zone anti-windup PID with higher gains ($K_p = [600, 1000, 450]$). Layer 3 adaptively re-weights the two layers using an inverse-error confidence metric over a rolling 48-hour window (EMA learning rate 0.01, initial weights [0.60 MPC, 0.35 PID, 0.05 safety]), with an explicit rule that shifts weight toward PID when MPC persistently under-predicts. A bounded proportional safety term, capped at 40% of rated capacity, guards against fusion failure.

3.5 Simulation Environment and Evaluation Metrics

All models run as MATLAB Function blocks inside a single Simulink model (`Pune_HVAC_Complete.slx`), comprising per-zone thermal subsystems, five switchable controller subsystems, the coupled VRF/IEC/DOAS equipment subsystem, and the IAQ subsystem, all driven by From-Workspace time-series blocks built from the TMY and building-parameter datasets. Each controller runs the full 8760-hour year at an hourly time step. The evaluation metrics: annual energy by end use (kWh); comfortable, cooling, and heating-needed hour counts; RMSE of Zone-2 temperature from the 24 °C set point; annual electricity cost under a three-tier Maharashtra ToD tariff (INR 5.20/kWh off-peak, INR 7.15/kWh normal, INR 8.95/kWh peak) [15]; annual embodied carbon (0.82 $kgCO_2$/kWh, Maharashtra CEA 2023 grid factor) [16]; and specific energy consumption (kWh/m^2).

4 Results

4.1 Zone Temperature Response and Set-Point Tracking

Fig. 1 shows the full-year zone-temperature trajectories for all five controllers against outdoor temperature and the 22–26 °C comfort band, for Zone 2 (top) and Zones 1 and 3 (middle, bottom). The baseline shows severe, sustained excursions in both directions. Cross-referencing the simulation's outdoor-temperature record confirms that hours around 2200–4300, the interval of the baseline's deepest undershoot, fall in Pune's hottest pre-monsoon period (mean outdoor temperature 34.1 °C, peaking at 47.4 °C), not a cool season. The baseline's cooling command has no deadband, anti-windup, or rate limiter (Section 3), so a single hourly timestep of near-maximum commanded cooling during this hot, high-error period can remove more heat than Zone 2's thermal capacitance (14.90 MJ/K, Table 1) can absorb without a large overshoot. With no hysteresis to prevent immediate re-triggering, this produces a sustained bang-bang undershoot/recovery cycle that drives Zone 2 down to roughly 8–10 °C for about 2000 hours, until the plant's cooling demand naturally subsides with the onset of the monsoon. All four advanced controllers, each equipped with a deadband and/or rate limiter, keep every zone within a narrow band close to the 24 °C set point throughout the same period.

Fig. 2 quantifies this as signed tracking error from the 24 °C set point. The baseline's RMSE of 6.84 °C is dominated by the hot-season overshoot/undershoot cycle just described, with error reaching −16 °C, plus a separate warm-season overshoot beyond +5 °C. All four advanced controllers cut RMSE by 68–75%. MPC (1.734 °C) and Hybrid (1.726 °C) round to the same 1.73 °C and are statistically indistinguishable, with Hybrid marginally ahead. PID follows closely at 1.79 °C. RL-DQN, though far better than the baseline, is the least accurate of the four at 2.17 °C, consistent with its coarse six-level discrete action space and 1.0 °C no-action deadband.

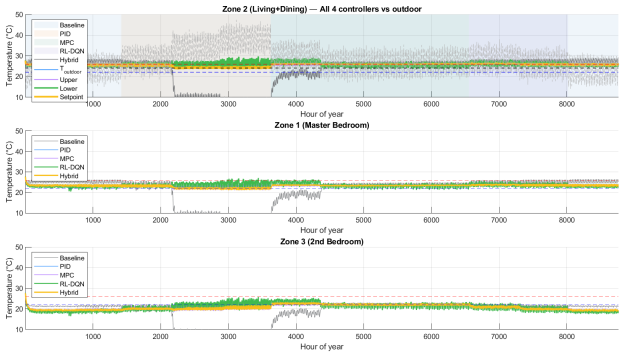


Fig. 1 Annual zone-temperature trajectories for all five controllers vs. outdoor temperature, comfort bounds, and set point — Zone 2 (top), Zone 1 (middle), Zone 3 (bottom).

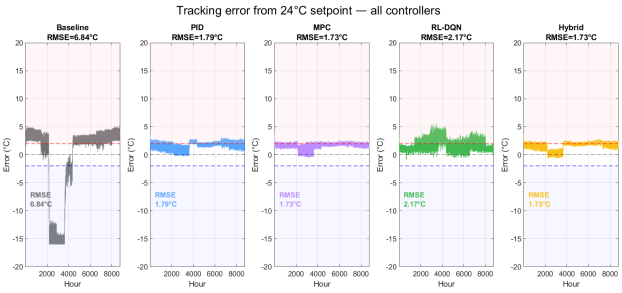


Fig. 2 Zone-2 tracking error from the 24 °C set point for all five controllers, with RMSE annotated per panel.

4.2 Comfort Distribution

Table 2 summarizes tracking accuracy and comfort-hour distribution. Fig. 3 (left) breaks annual hours into heating-needed (< 22 °C), comfortable (22–26 °C), and cooling-needed (> 26 °C) bands for Zone 2. The baseline reaches comfort in only 11.3% of hours. All four advanced controllers essentially eliminate the heating-needed band and raise the comfortable-hour fraction six to seven-fold: MPC is strongest at 72.8%, PID and Hybrid sit tied near 68%, and RL-DQN trails at 57.8%. The temperature-duration curve (Fig. 3, right) confirms this ordering. The baseline's curve falls to about 8 °C at its lower tail, while the four advanced controllers cluster tightly around the comfort bounds.

4.3 Energy Consumption

Fig. 4 (left) shows monthly energy consumption. The baseline uses 580–605 kWh every month, reflecting inefficient, poorly timed duty cycling, while all four advanced controllers cluster between roughly 430 and 495 kWh/month. Fig. 4

Table 2 Tracking Accuracy and Comfort-Hour Distribution

Controller	RMSE (°C)	Comfort. (%)	Cooling (%)	Heating (%)
Baseline	6.84	11.3	65.7	23.0
PID	1.79	68.2	31.8	0.0
MPC	1.73	72.8	27.2	0.0
RL-DQN	2.17	57.8	42.2	0.0
Hybrid	1.73	68.1	31.9	0.0

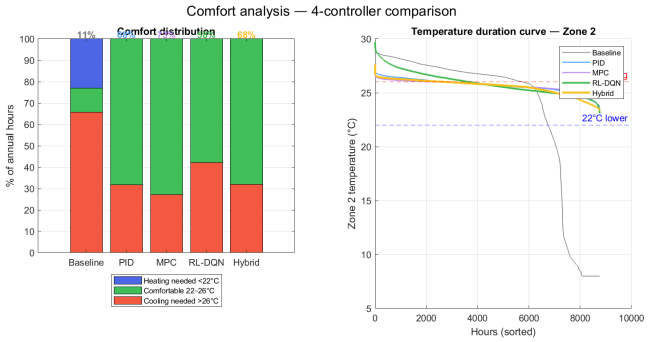


Fig. 3 Comfort-band distribution (left) and Zone-2 temperature-duration curve (right) for all controllers.

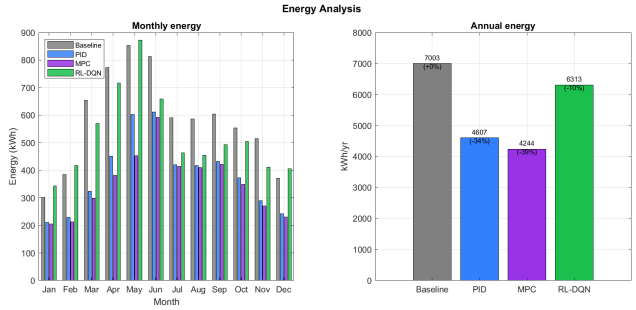


Fig. 4 Monthly energy consumption for all controllers (left) and annual energy decomposed by VRF, IEC, and DOAS end use (right).

(right) and Table 3 break annual energy down by end use. Baseline VRF energy (5816.3 kWh, 81.8% of its 7110.8 kWh total) reflects that overshoot/undershoot cycling. All advanced controllers cut VRF energy to 4166–4682 kWh.

PID achieves the lowest total annual energy, 5460.9 kWh, a 23.2% cut versus baseline, mainly through the lowest VRF draw (4166.4 kWh), a result of tight per-zone deadbands and anti-windup logic. Hybrid follows closely at −21.8%. MPC and RL-DQN save less, −18.7% and −18.4%, despite MPC's superior tracking, because MPC's anticipatory cooling draws more VRF energy to hold its tighter band. DOAS energy (920.5 kWh/yr) is identical across controllers since it is purely weather-triggered. IEC energy ranges from 202.8 kWh (RL-DQN) to 374.0 kWh (baseline, PID).

Table 3 Annual Energy Consumption by End Use (kWh)

Ctrl.	Total	VRF	IEC	DOAS	Δ vs. base	Spec.
Baseline	7110.8	5816.3	374.0	920.5	—	74.85
PID	5460.9	4166.4	374.0	920.5	−23.2%	57.48
MPC	5779.5	4623.3	235.6	920.5	−18.7%	60.84
RL-DQN	5805.0	4681.7	202.8	920.5	−18.4%	61.11
Hybrid	5558.8	4297.3	341.0	920.5	−21.8%	58.51

Table 4 Annual Electricity Cost and Embodied Carbon

Ctrl.	Cost (INR)	Δ cost	Carbon ($kgCO_2$)	Δ carbon
Baseline	50,842	—	5,830.9	—
PID	39,045	−23.2%	4,477.9	−23.2%
MPC	41,323	−18.7%	4,739.2	−18.7%
RL-DQN	41,506	−18.4%	4,760.1	−18.4%
Hybrid	39,745	−21.8%	4,558.2	−21.8%

Cost and carbon are linear functions of the same energy series (fixed ToD tariff [15] and grid factor [16]), so percentage savings in cost and carbon track energy savings exactly (Table 4). Switching from baseline to PID or Hybrid saves INR 11,000–12,000/yr in electricity cost and 1.05–1.35 tCO_2/yr in embodied emissions, for this single dwelling.

4.4 Time-of-Day Load Shifting and VRF Performance

Fig. 5 (left) shows mean hourly VRF power against the three-tier ToD tariff. Hybrid and PID hold the flattest, lowest hourly power profile, about 490–495 W, largely uncorrelated with the tariff, while MPC and RL-DQN sit modestly higher, around 528–536 W. None shows pronounced load-shifting away from the 07:00–09:00 and 19:00–23:00 peak-tariff windows toward the 00:00–06:00 off-peak window. As Section 3 established, MPC's absence of load shifting comes entirely from its tariff-weighting term never connecting to its output. It is simply not tariff-aware in practice. Hybrid's case is different. Its tariff multiplier (0.88–1.35× on the predictive-layer demand, favoring off-peak cooling) is genuinely wired into the control law, yet it produces no visible flattening or off-peak increase in the aggregate hourly profile. That suggests the effect gets masked by the controller's staged temperature-error overrides and by how much of the fused output the PID and safety layers claim, rather than by any absence of a tariff-sensing mechanism.

Fig. 5 (right) shows the modeled VRF COP curve, falling from about 5.3 at 28 °C to 4.2 at the 35 °C ISEER condition [9], to about 3.4 at Pune's documented 41 °C peak, and to about 3.0 at the most extreme 44 °C condition modeled, a 43% efficiency drop across that range. That explains why

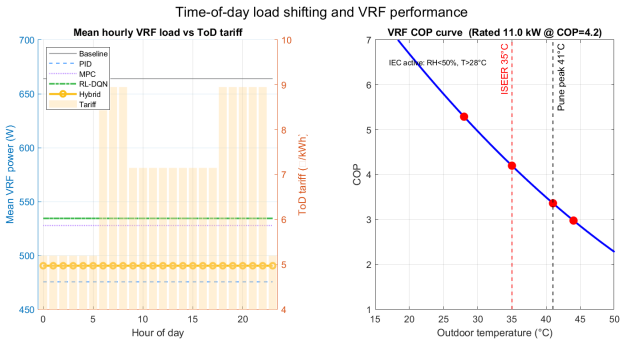


Fig. 5 Mean hourly VRF power vs. time-of-day tariff for all controllers (left) and the VRF COP curve vs. outdoor temperature, with IEC-activation and ISEER/Pune-peak reference points (right).

controllers avoiding unnecessary operation during the hottest hours see disproportionate energy savings.

4.5 Indoor Air Quality

Fig. 6 (left) shows the annual indoor CO_2 profile, identical across all controllers since DOAS activation depends only on outdoor weather. CO2 concentrations remained around 900–920 ppm during the dry seasons, when the DOAS was inactive. During the monsoon period, approximately hours 3600–6600, concentrations fell to around 420–600 ppm as outdoor relative humidity exceeded 65% and the DOAS began supplying fresh air[10]. That shows humidity-triggered ventilation works as an effective, if incidental, CO_2-control strategy. The dashed reference line in Fig. 6 (left) is an internal threshold used by the simulation model, not a value traceable to a specific ASHRAE 62.1/62.2 clause, and should be read as the model's own indoor-CO_2 reference rather than a certified standard limit.

Fig. 6 (right) tells a different story for particulate matter. Despite a 99.95%-efficient HEPA H13 filter, indoor $PM_{2.5}$ stays between about 160 and 195 µg/m^3 year-round, more than three times the Indian NAAQS 24-hour limit of 60 µg/m^3 [18] and more than ten times the WHO 24-hour guideline of 15 µg/m^3 [19]. Like CO_2, it is identical across controllers, because it is governed by the fixed 30% recirculation/HEPA architecture and the outdoor $PM_{2.5}$ series, not by the thermal controller. Only the recirculated fraction of air passes through the HEPA filter, so unfiltered infiltration keeps reintroducing outdoor particulate pollution [11,17], a limitation of recirculation-only filtration reported in other outdoor-$PM_{2.5}$ filtration studies [11]. The model's $PM_{2.5}$ balance gives DOAS fresh air no explicit ingress pathway, so the exceedance shown here traces back to infiltration and the recirculation-only filter architecture, not to DOAS operation.

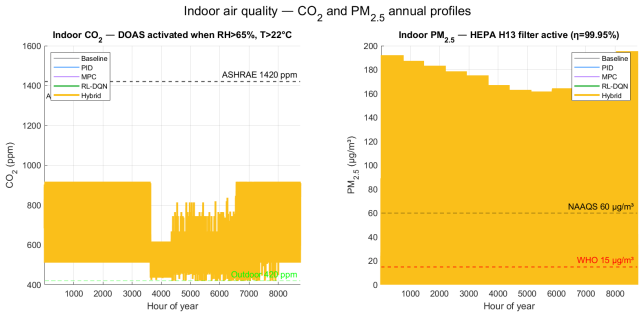


Fig. 6 Annual indoor CO_2 profile with DOAS activation (left) and indoor $PM_{2.5}$ profile with HEPA H13 filtration (right), against ASHRAE, WHO, and Indian NAAQS reference levels.

5 Discussion

The results lay out a clear, three-way trade-off among tracking accuracy, energy use, and controller complexity. MPC (1.734 °C) and Hybrid (1.726 °C) are statistically indistinguishable on tracking, but Hybrid gets there with meaningfully lower energy, 5558.8 versus 5779.5 kWh/yr, because its PID layer and adaptive fusion weighting avoid MPC's more aggressive anticipatory cooling. The plain PID controller, despite being the simplest of the four advanced options, ends up the most energy-efficient at 5460.9 kWh/yr, sacrificing only marginal tracking accuracy. That result cuts against the assumption that more sophistication automatically buys proportionally better outcomes, and it lines up with recent experimental comparisons showing that well-tuned classical controllers can match or approach advanced-control energy performance over long evaluation periods [26].

We suspect, though this study does not include the ablation needed to confirm it, that RL-DQN's weaker showing traces back to its deployment protocol: a pretrained policy evaluated greedily, a coarse six-level action space, a wide no-action deadband, and, per Section 3, a night-setback-blind state representation, with no further learning against this building's dynamics during the evaluation year. That reading fits reports elsewhere that RL policy performance is highly sensitive to the match between training and deployment conditions and to opportunities for online adaptation [20,21,22,24], and that finer action resolution and occupant- or schedule-aware state design have improved DRL controller performance in other studies [23,25,27]. We offer this as a plausible explanation, not a demonstrated cause, and flag it as a direction for follow-up work in Section 6.

The Hybrid controller is based on the idea that combining predictive and reactive control can take advantage of both approaches. The results support this idea to a reasonable extent. Hybrid achieved tracking performance comparable to MPC while coming closer to PID in terms of energy efficiency, making it a balanced option among the controllers evaluated in this study.

The load-shifting analysis in Section 4 helps explain why neither MPC nor Hybrid achieved substantial changes in the load profile. In the MPC implementation, the tariff term is disconnected from the final control output, so it does not influence the controller's decisions. The Hybrid controller includes a functional tariff term, but its effect on the load profile appears limited because temperature-error overrides and fusion weighting take priority. These findings highlight the need to verify that tariff-based control signals influence the final system output, rather than checking only whether they appear in the cost function. They also point to an opportunity to develop a more effective pre-cooling strategy that takes advantage of the building's long thermal time constants of 56–66 hours. The IAQ results also show that achieving thermal comfort does not necessarily lead to healthier indoor conditions. CO_2 and $PM_{2.5}$ followed identical trajectories across all five controllers, indicating that indoor air quality was determined mainly by the fixed ventilation and filtration arrangements rather than by the thermal control strategy. $PM_{2.5}$ remained high despite the use of a highly efficient filter. This suggests that filtration based only on recirculated air is insufficient to control outdoor particulate matter entering through infiltration, particularly under Pune's ambient pollution conditions.That matches findings elsewhere that outdoor-air filtration strategy, not filter efficiency alone, determines indoor $PM_{2.5}$ outcomes [11,17]. It points to filtering incoming air, both infiltration and ventilation, not just recirculated air, as a necessary complement to any control strategy evaluated here.

6 Limitations

- **Single building, single climate:** findings are specific to this 95 m^2, three-zone Pune dwelling and its TMY-derived weather profile.
- **Night-setback asymmetry:** Baseline and RL-DQN regulate against a flat 24 °C target, while PID, MPC, and Hybrid apply a 2 °C night setback. This is a genuine, partial confound in the RL-DQN versus PID/MPC/Hybrid comparison that we do not fully separate from algorithmic performance differences.
- **Simplified control-internal models:** MPC and Hybrid use low-order, first-order predictive models for real-time tractability, not the full multi-zone RC network used for the plant.
- **Static, untested RL policy:** RL-DQN was evaluated frozen, in greedy inference mode, with no online adaptation to the specific evaluation year. We did not run the ablations (finer action resolution, online fine-tuning, setback-aware state features) needed to confirm which factor or factors drive its underperformance. The explanation offered in Section 5 is a hypothesis, not a demonstrated result.
- **No latent-load or humidity comfort metric:** comfort is scored on dry-bulb temperature only.

- **Fixed occupancy:** occupancy and internal gains use fixed schedules for an assumed four-occupant household rather than measured or stochastic behavior. No occupant-override loop is modeled.
- **$PM_{2.5}$ scope:** the finding is specific to the modeled recirculation/HEPA configuration and outdoor $PM_{2.5}$ series. The model does not give DOAS ventilation air an explicit $PM_{2.5}$ ingress pathway. A model that did would likely show a larger exceedance, not a smaller one.
- minor inconsistency exists between the building model and the IAQ subsystem. The building-envelope model uses an infiltration rate of 0.0376 m^3/s, based on a 0.5 ACH design assumption, while the IAQ subsystem uses a hardcoded value of 0.0375 m^3/s The difference is less than 0.1% and has no meaningful numerical impact, but it is documented here for reproducibility.
- **Economic scope:** results use a single, fixed ToD tariff and a single grid-carbon-intensity factor. Sensitivity to tariff structure or grid decarbonization is out of scope.

7 Conclusion

This study ran a year-long, physics-based Simulink simulation comparing five HVAC control strategies, baseline rule-based, PID, MPC, RL-DQN, and a Hybrid MPC–PID–safety fusion controller, on an identical three-zone 3BHK residence, VRF/IEC/DOAS equipment set, and Pune climate profile. All four advanced controllers dramatically outperformed the reactive baseline: they cut RMSE by 68–75%, raised comfortable-hour fraction from 11.3% to 58–73%, and reduced energy, cost, and carbon by 18–23%. Overall, PID achieved the greatest energy savings, MPC and Hybrid delivered comparable tracking performance, and Hybrid offered a balanced trade-off between tracking accuracy and energy consumption. The greedily deployed RL-DQN policy lagged the other three, which underlines that deployment protocol matters as much as algorithm choice in RL-based building control.

Evaluating air quality alongside thermal performance showed that CO_2 and $PM_{2.5}$ outcomes were entirely insensitive to the choice of thermal controller. They were governed instead by the fixed ventilation and filtration architecture, with indoor $PM_{2.5}$, driven by infiltration and a recirculation-only HEPA filter rather than by DOAS operation, persistently exceeding NAAQS and WHO guidelines. Together, these results argue for hybrid, adaptively weighted control as a practically attractive middle ground in residential HVAC. They also argue for deliberate pre-cooling and load-shifting design if ToD tariff exposure is to be meaningfully exploited, and for treating filtration architecture as a first-class design variable alongside control strategy whenever indoor air quality is a design objective.

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